



DEALER BEST PRACTICE SERIES

Bulk Oil Filtration

Application Maintenance Site Management Component Rebuild Component Life Management MARC Management

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1.0 Introduction

Improving component life is a key factor in achieving the lowest cost per ton. Few factors can extend component life as much as clean oil.

Modern powershift transmissions and implement hydraulic systems use large numbers of electro-hydraulic control valves, which require very clean oil to operate properly. Even in gear-case environments, such as the final drives and differentials, roller bearings are highly loaded and can be damaged from microscopic abrasive debris in the oil.

Fluid cleanliness begins by ensuring that oil put in the machine is clean to begin with.

Nearly all new oil is contaminated to some degree with dirt, metal particles, plastic, water, or other foreign debris. A widespread misconception is that new oil is automatically clean because of its appearance. However, contaminants are introduced in the transportation and storage process from the time the oil leaves the refinery until it is used. Very little new bulk oil meets the recommended Caterpillar cleanliness spec for new oil of ISO16/13. This includes oil delivered in bulk tanks, steel barrels, plastic cubes, and small plastic containers.

New oil that does not meet the cleanliness spec should be filtered until it does meet the following cleanliness specs:

Desired	ISO 16/13
Maximum	ISO 18/15

There are a variety of methods for filtering new oil. These include:

1. Permanent filtration installation
2. Portable filtration
 - a. Barrels
 - b. Cubes
3. On-Machine

The method used may vary depending on a number of factors. These include:

1. Volume of oil being used
2. Location
3. Available infrastructure
4. Cost

2.0 Best Practice Description

2.1 Permanent Filtration Installation

Large, permanently installed filtration systems are often the best choice for facilities which use large volumes of new oil and have centralized oil supply systems.

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Typical users of permanent filtration systems include:

- Dealer main store shops
- Large branch stores
- Component rebuild centers
- Large shops at mine sites.

Permanent systems usually consist of three major components:

2.1.1 Main Storage Tank

A large storage tank holds the entire supply of a particular grade of oil, and usually has a capacity of several thousand gallons. A large storage tank is required for each grade of oil used.

2.1.2 Day Tank

It is time-consuming and impractical to try to clean several thousand gallons of oil in the main storage tank. The primary reason is that the entire supply will be re-contaminated with unfiltered oil when the tank is re-filled. The recommended method is to pump oil from the storage tank into a smaller day tank sized, which has adequate capacity to supply needs for 1-2 days. The smaller volume of the day tank can be refilled and filtered at night for use the next day. It is also not affected by re-filling of the main storage tank.

2.1.3 Re-circulating Filtration System

Permanently mounted pumps continuously circulate oil from the day tank, through a bank of particulate filters and back into the day tank. This process is usually performed at night during off-shift periods continues for several hours until the desired cleanliness is achieved in the day tank. While this system does require a sizeable capital investment, it is the most economical method for supplying large volumes of clean oil for high volume applications.

Caterpillar does not supply permanent system components, but can assist in recommending proper system design and reputable suppliers.

2.2 Portable Filtration

Large, permanently installed filtration systems are often too expensive for smaller branch stores and mine site repair shops. In these applications, various forms of portable filtration is often the best option.

2.2.1 Filter Carts

Smaller facilities often purchase and store new oil in steel barrels or 1,000-liter plastic cubes. Caterpillar offers portable filter carts and fittings to clean the oil in the barrels and cubes before it is pumped into the machine. This is the same basic process of re-circulating oil through a filter bank as permanent installations, except on a much smaller scale. Portable carts are much less expensive, but also filter much smaller volumes and are more labor intensive to manage. However, they are the most practical and economical alternative for smaller facilities.

2.3 On-Machine Filtration

In some application, the volume of new oil used is inadequate to justify permanent filter systems, but enough to make filtering individual cubes and barrels difficult to manage. While it is desirable

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to clean new oil before it is pumped into the machine, new oil may also be cleaned after it is placed in the machine. This is a two-step process:

2.3.1 Series Filtration from the Container

Unfiltered new oil should never be taken directly from the container and placed into the machine. This is true whether the oil is being used for refilling a compartment at an oil change interval or simply topping off a system.

If a re-circulating filter system is not used, a simple single-pass filter arrangement offers an acceptable alternative. The filter arrangement consists of three 5 micron, beta 75 ultra-high efficiency hydraulic filters in series followed by a water absorbing filter. The new oil flows from the container through each of the five-micron filters in series, and then through the water-absorbing filter.

The filter bases must not contain bypass valves, so that if particulate filters begin to plug, oil flow stops requiring the plugged filters to be changed. The water-absorbing filter effectively absorbs any water in the oil and automatically blocks flow when the water absorbing media is depleted.

Using this arrangement will assure that most large particulate contaminants and all excess water is removed from the new oil.

2.3.2 Off-Board Machine Filtration (Kidney Loop)

The use of off-board filtration carts is required to clean machine systems after a major component or system repair before the machine is returned to service. They are also commonly used during machine PM to clean the oil while the machine is already down for service. These carts should also be used during oil changes to filter new oil that was not cleaned to ISO 16/13 before being placed in the machine.

The new oil must be filtered in the compartment until it reaches the desired cleanliness before the machine is started or placed backing service.

3.0 Implementation Steps

3.1 Permanent Filtration

System design guidance and supplier recommendations can be obtained from the Caterpillar Service Tools Group or from the Caterpillar Product Support Division Contamination Control Group.

3.2 Portable Filtration

The size and number of portable filtration carts needed for filtering new oil containers is determined by the number of different oil types and type of containers being used. A wide variety of carts are available from Caterpillar. Information may be obtained through the Contamination Control Group or Service Tool Group.

3.3 On-Machine Filtration

Series filter arrangements for each grade of oil is required. These are an inexpensive method to fabricate using filter bases and UHE hydraulic filters available through the Caterpillar parts

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system. Caterpillar no longer offers water absorbing filter elements. These may be easily obtained from a number of large filter manufacturers. Part numbers of UHE filters and filter bases are available through the Caterpillar Fluids and Filters group.

Off-board filtration carts are already required for major system and component repairs. They are also commonly used during PM. There are a variety of filter cart sizes available through Caterpillar. Information may be obtained from the Caterpillar Service Tools Group or from the PSD Contamination Control Group.

4.0 Benefits

Assuring clean oil is supplied to machine systems **increases both reliability and durability.**

Electro-hydraulic control valves, which are widely used in transmission controls and implement hydraulics, are very intolerant of microscopic, ferrous debris.

Heavily loaded wheel and final drive bearings, as well as, duo-cone seals are also easily damaged by abrasive debris commonly found in new oil.

Assuring the new oil is cleaned to the desired cleanliness level **significantly reduces the number of contamination-induced failures and repairs.**

5.0 Resources Required

See implementation steps above.

6.0 Supporting Attachments / References

An explanation of the various methods of bulk oil filtration, along with visuals is available in the [Improving Component Durability -Managing Fluid Cleanliness](#) booklet.
(Form Number [SEBF1020](#))

7.0 Related Best Practices

See also:

0806-2.10-1000 -Fluid Cleanliness Management
0806-2.10-1006 -Bulk Fuel Filtration

8.0 Acknowledgements

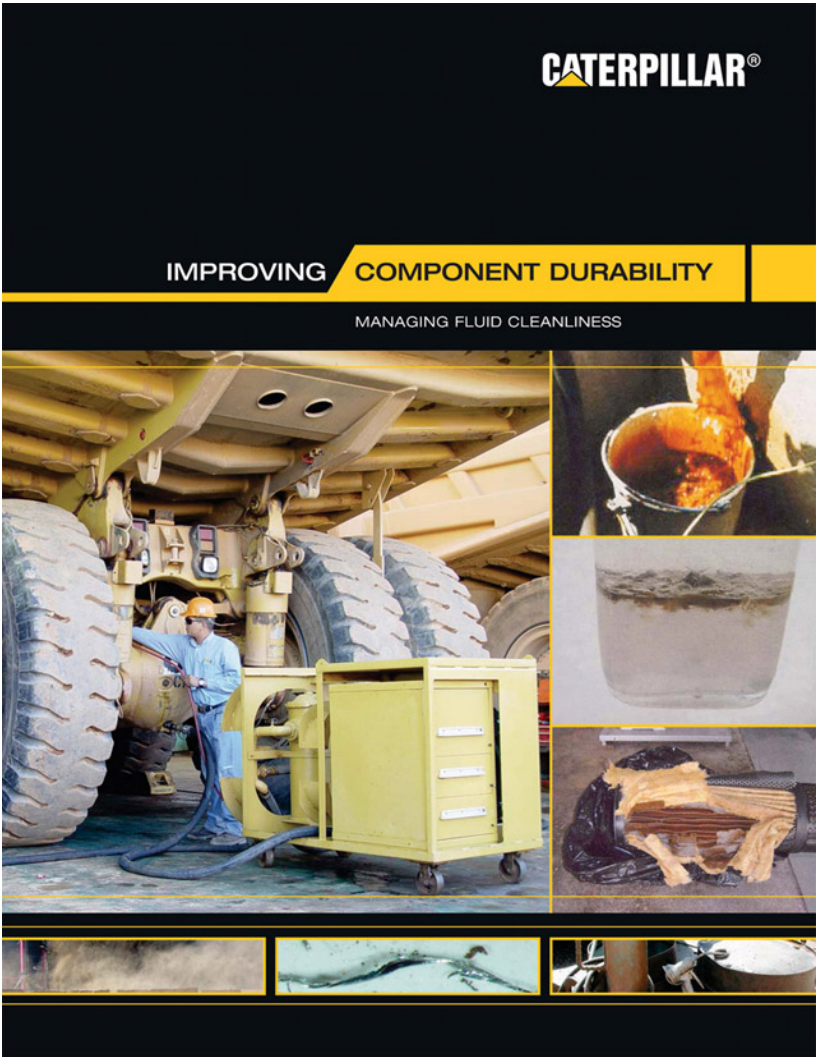
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Improving Component Durability -Managing Fluid Cleanliness booklet.
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